



EDUCATION			
YEAR	DEGREE / EXAM	INSTITUTE	CGPA / MARKS
2023-2027	Bachelor of Technology (Chemical Engineering)	Indian Institute of Technology Tirupati	8.55
2022	Senior Secondary	Tamil Nadu State Board	92.5%

EXPERIENCE

Material Processing Intern | e-TRNL Energy May '26 - July '26
Tools & Technologies: Battery Electrode Manufacturing, Semi-Dry Mix (SDM) Processing, Calendering, Tap Density & Porosity Testing, N/P Ratio Matching, Lithium-ion Coin Cell Assembly

- Synthesized** Semi-Dry Material (SDM) batches for 3D Electrode Architecture (3DEA) anodes and punched out the 3DEA anodes.
- Performed** daily testing on powders and electrodes to measure tap density and porosity to ensure excellent batch consistency.
- Proposed** a new measurement approach for the lab's testing protocols across different electrode shapes, reducing overall testing time.
- Operated** full assembly lines for lithium-ion coin cells; coated SDM onto current collector sheets, pressed them flat under heavy rollers (calendering), and punched out neat electrode shapes to achieve desired porosity.
- Built** Python automation scripts to match battery electrodes by capacity (N/P ratio) for 6-electrode cells, and to convert raw battery tester data into clean performance graphs.
- Observed** the complete pouch cell and 3DEA electrode manufacturing process firsthand.

Research and Development Intern | Avkalan Labs Dec '25 - Apr '26
Tools & Technologies: Python, Finite Difference Method (FDM), Single Particle Model (SPM), Physics-Informed Neural Networks (PINNs), Battery Management Systems (BMS), Lithium-ion Battery Modeling

- Improved** the PINNSTRIPES neural network codebase so it could accurately predict battery voltage during dynamic, rapidly changing charge and discharge cycles.
- Built** a custom Single Particle Model (SPM) numerical solver from scratch (using FDM for solid-state diffusion) to accurately output cell voltage and internal concentration profiles under real-world, dynamic BMS current loads.
- Simulated** real BMS operating conditions for a 50Ah prismatic cell using a custom SPM solver, achieving a maximum voltage prediction error of 80 mV.

Research Intern | Robell MRC May '25 - Jun '25
Tools & Technologies: Hydrothermal Synthesis, Cyclic Voltammetry (CV), Galvanostatic Charge-Discharge (GCD), Zn-ion Batteries

- Synthesized** K⁺/Cu²⁺ co-doped Mn₂O₄ cathode materials via hydrothermal co-precipitation for aqueous Zn-ion batteries.
- Achieved** a specific capacitance of 201.5 F/g and an energy density of 31.7 Wh/kg at 1 A/g, while reducing internal resistance by 50% compared to the baseline.
- Investigated** charge storage kinetics using cyclic voltammetry, revealing a mixed pseudocapacitive-diffusion-controlled mechanism with up to 56% capacitive contribution.

PROJECTS

Simulation-Guided Optimization of 3D-Printed Electrode Architecture for High Loading Sodium-Ion Batteries (Ongoing)
Tools & Technologies: Python, 3D CAD, Electrochemical Modelling, Direct Ink Writing (DIW), Rheology, EIS, GCD & Battery Cycling

- Developing** 3D-printed sodium-ion battery electrodes using hard carbon and Na₃V₂(PO₄)₃ (NVP), engineering architected porosity to enhance Na⁺ transport, active-material utilization, areal capacity, and rate capability at high electrode loading.
- Modeling** the effect of pore size and 3D electrode architecture on ionic transport to identify the optimal design, validating it using DIW 3D printing and electrochemical testing.

Physics-Based Multi-Stage Constant Current (MCC) Fast-Charging Optimization & Lithium-Plating Risk Modeling for Aged Li-ion Cells | GitHub

Tools & Technologies: Python, PyBaMM, Doyle-Fuller-Newman (DFN) Model, Differential Evolution, SciPy, Lithium-ion Battery Degradation Modeling

- Built** a high-fidelity electrochemical simulation of an aged LG INR21700-M50T Li-ion cell using PyBaMM's DFN model, calibrated to real degradation test data (capacity and internal resistance across multiple state-of-health checkpoints from lab RPT measurements).
- Optimized** a 4-stage MCC charging protocol via differential evolution, achieving 24-26% faster charging than baseline CC-CV across a 70-100% SOH aging sweep while holding a 20mV lithium-plating safety margin, validated via a 4x larger optimizer search budget (results agreed within ~1%).
- Extended** limited real aging measurements (88-91% SOH) into a full 70-100% SOH degradation sweep using a physics-informed resistance model anchored to real capacity/resistance data and literature-reported degradation trends.
- Extended** the MCC optimizer to jointly tune stage-transition SOC boundaries alongside stage currents, recovering an additional 3-5% charging-time improvement over a fixed-staging baseline.

Multi-Stage Constant Current Fast-Charging & Thermal Simulation for NMC Li-ion Battery Pack | GitHub

Tools & Technologies: MATLAB, Simulink, Simscape Battery, Stateflow, Equivalent Circuit Modeling (2RC ECM)

- **Designed** and simulated a multi-stage fast-charging algorithm for a 4-cell lithium-ion battery pack in MATLAB/Simulink, achieving 80% charge ~12.5 minutes faster than standard CC-CV charging while enforcing strict safety limits (4.20V, 60°C).
- **Built** a coupled electro-thermal battery model (2RC equivalent circuit + Stateflow control logic) capturing dynamic voltage response and cell-to-cell heat conduction across a 4-cell series pack.
- **Developed** and compared two active thermal management strategies (air vs. liquid cooling) using hysteresis-based relay control, showing liquid cooling reduced peak cell temperature by ~5°C during high-current charging.
- **Validated** system safety and performance through full-cycle simulation, confirming zero voltage violations and stable thermal regulation across all charging stages.

Physics-Based Thermal Modelling of a 21700 Li-ion Cell using PyBaMM (Ongoing)

Tools & Technologies: Python, PyBaMM, Doyle-Fuller-Newman (DFN) Model, Electrochemical Battery Modeling

- **Benchmarking** a published 3D COMSOL electro-thermal study via open-source Python (PyBaMM) to simulate and validate transient temperature rise and heat generation profiles for high-energy 21700 Li-ion cells.
- **Implemented** a high-fidelity Doyle-Fuller-Newman (DFN) lumped-thermal model; optimized spatial mesh resolution and CasADi solver configurations to resolve stiff diffusion equations and accurately quantify ohmic, reversible, and irreversible heat fractions.

TECHNICAL SKILLS	
Battery & Electrochemical Modeling	DFN/P2D, SPM, Equivalent Circuit Modeling (1RC/2RC ECM), Electro-Thermal Simulation, Degradation & Li-Plating Modeling
Numerical Methods & Optimization	FDM, Differential Evolution, CasADi, Parameter Estimation
BMS & Battery Controls	MCC/CC-CV Fast-Charging, SoC & SoH Estimation, Thermal Management (Air/Liquid Cooling), Stateflow Control Logic
Simulation & Programming	Python (PyBaMM, NumPy, Matplotlib), MATLAB/Simulink/Simscape Battery, Git/GitHub
Electrode Manufacturing & Characterization	Li-ion Coin & Pouch Cell Assembly, SDM Processing & Calendering, Tap Density & Porosity Testing, N/P Ratio Matching, EIS, CV, GCD

KEY COURSES TAKEN	
Battery Manufacturing Technology Electrochemical Energy Storage Computational Techniques for Chemical Engineers Chemical Reaction Engineering Process Control and Dynamics Introduction to Renewable Energy	

POSITIONS OF RESPONSIBILITY	
• Placement Representative, Chemical Engineering (B. Tech) Jun'26 - Present • Captain, IIT Tirupati Athletics Team Aug '25 – May '26 • Co-Coordinator, Event Management, Chemical Engineering Club, IIT Tirupati Jul '26 - May '26 • Course Project Group Head, Ecology course Project (a group of 15 members), IIT Tirupati Jan'25 - Apr '25 • The Vice Captain, Dragoons Sports team, Kridhan-2025, IIT Tirupati Mar '25- Apr '25	

CERTIFICATIONS	
• Battery and Battery Management Systems – DIYGURU, July 2025 • Battery Pack Modeling – MathWorks, July 2025 • Battery State Estimation– MathWorks, July 2025 • Simscape Battery– MathWorks, July 2025	

ACHIEVEMENTS	
• 2nd Place – D'Solve It 24-hour Simulink Hackathon, Chemical Engineering Club, IIT Tirupati Nov. 2025 — Developed a pH process simulator and designed a feedback controller for pH regulation.	